AUTOMATION CASE STUDY

Maintenance request triage

From a tenant message to a reviewed request, approved vendor contact and a visible record of the outcome.

The problem

Maintenance messages arrive across channels with missing detail. Managers need a clear priority, the right trade and a reliable next step.

What I built

Five connected n8n workflows collect, clarify and organize requests, then put vendor contact behind a manager approval.

	Reference	Priority	Source	Trade
1	MR-0001	URGENT	telegram	PLUMBING
2	MR-0002	ROUTINE	email	APPLIANCE
3	MR-0003	ROUTINE	email	APPLIANCE
4	MR-0004	URGENT	form	PLUMBING
5	MR-0007	URGENT	telegram	PLUMBING
6	MR-0009	ROUTINE	form	APPLIANCE

Actual Airtable workspace with synthetic maintenance records.

The following pages show the build and a connected execution, rather than client performance claims.

ARCHITECTURE

Five workflows, shared state

n8n coordinates the work. Postgres retains ticket and conversation state; Airtable provides the manager view.

01	Tenant Intake	Email or form -> normalized request -> shared triage engine.
02	Triage Engine	Issue -> priority, trade and missing details -> saved ticket and eligible draft.
03	Tenant Telegram	Conversation -> clarification or factual status -> the same request.
04	Manager Desk	Vendor roster -> approval button -> email, tenant update and recorded result.
05	Operations	Scheduled checks -> emergency review and conversation sync to Airtable.

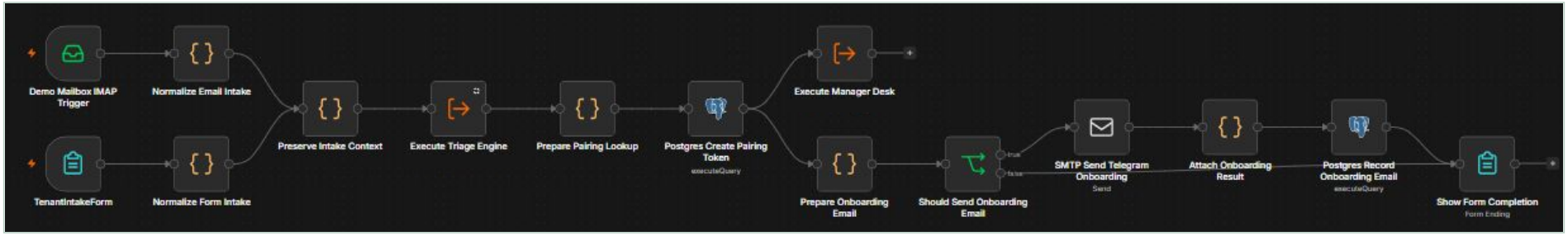
Demonstration scope

Demo Tenant, Maple Court, Unit 2B and the vendor names are synthetic. These captures use live OpenAI, Telegram, Airtable and Gmail through controlled demo accounts. The form was submitted in test mode. Vendor email went to a controlled demo mailbox, not a real contractor.

The visible fixture branch supplies mock classifications for rehearsal; the shown ticket journey used the live model.

Two entrances, one request

Email and the web form feed the same engine, preserving the source message and tenant context.



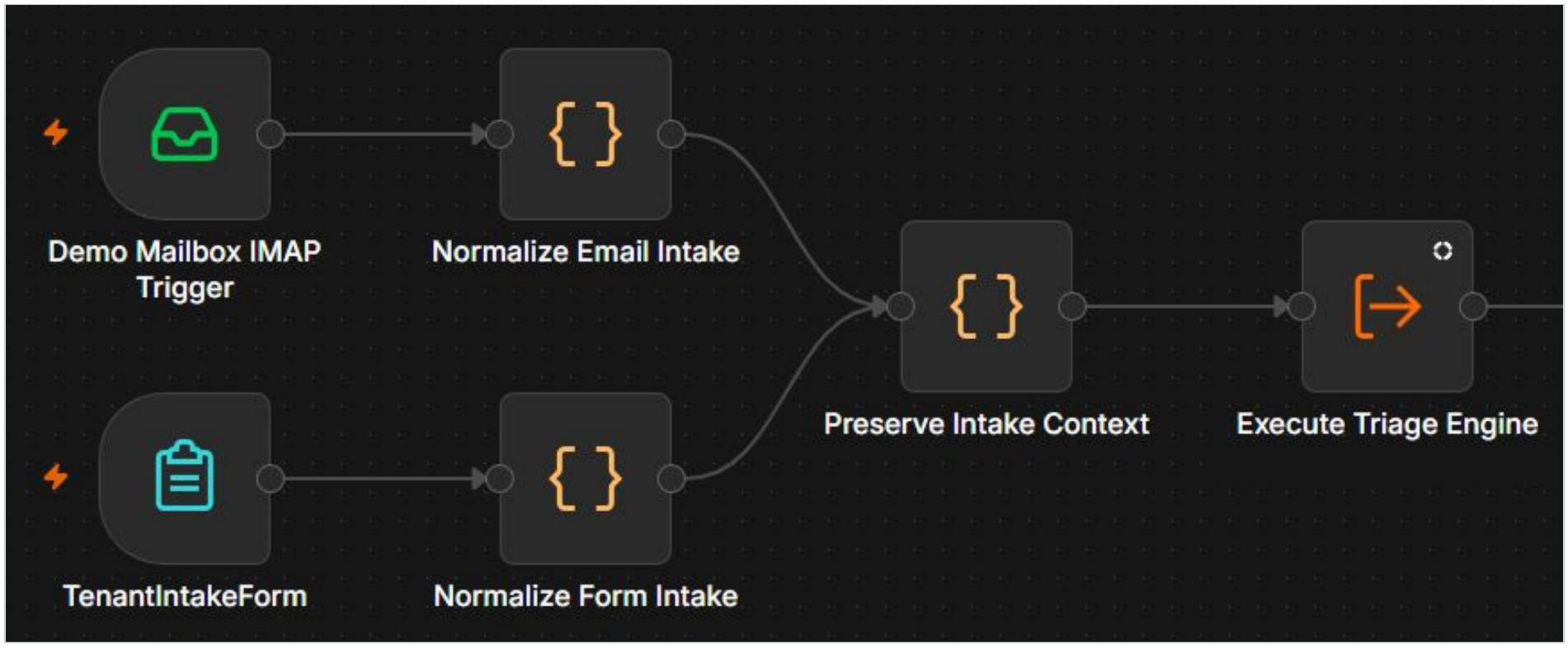
Actual n8n canvas: every node visible. Readable sections follow.

Trigger	An IMAP maintenance email or a form submission.
Incoming data	Issue text, property, unit, reply channel and entry permission when supplied.
Key steps	Normalize intake; call the engine; reselect its saved ticket; create a pairing token.
Output	A manager handoff when reviewable, a pairing email when eligible and a form receipt.
Approval	Vendor sending is handled later by the Manager Desk approval route.

WORKFLOW 01 / NORMALIZATION

Preserve the source before triage

Both entrances become one request shape with a stable source identifier.

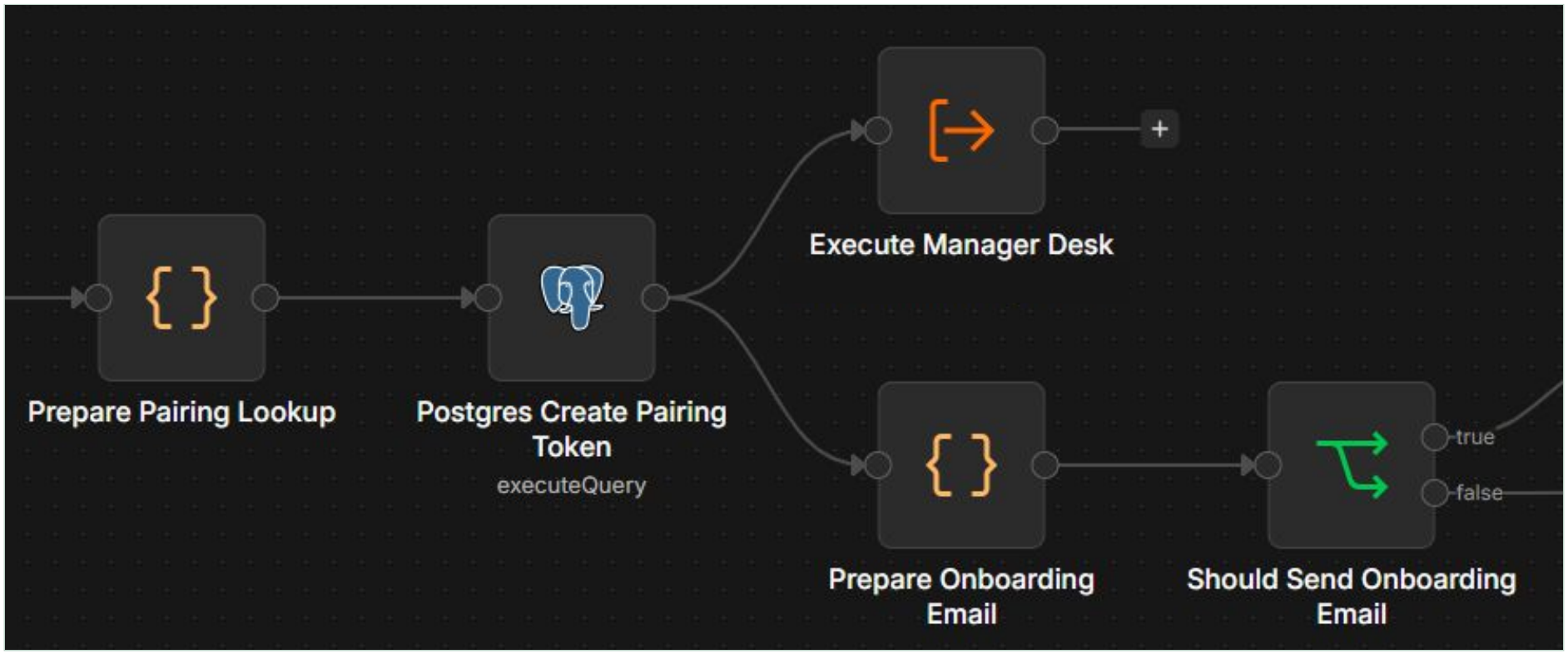


Actual editor capture: enlarged at native resolution.

The shared context retains the tenant message and its origin while the engine runs. Intake then reads the durable result before continuing. This avoids depending on a sub-workflow returning an item on every branch.

Connect the tenant and manager

The saved request controls two paths: manager review and tenant onboarding.

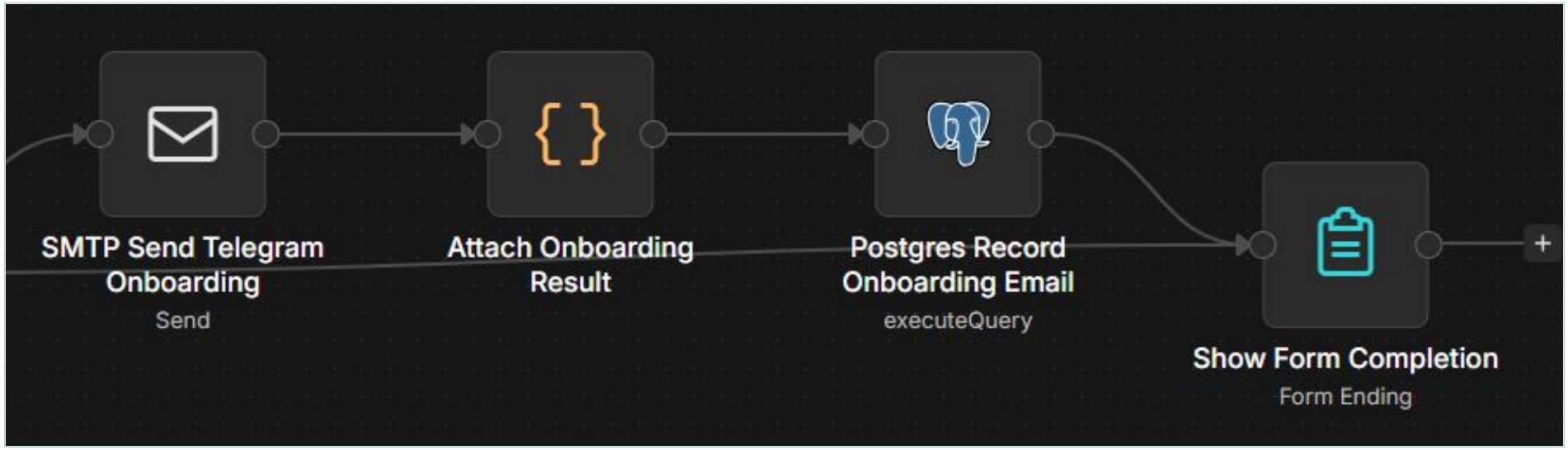


Actual editor capture: enlarged at native resolution.

A one-time token connects an email or form request to its Telegram conversation. Reviewable requests also enter the Manager Desk. The onboarding gate prevents an ineligible or repeated pairing email.

Record delivery, then finish

The onboarding message is recorded before the form displays its receipt.

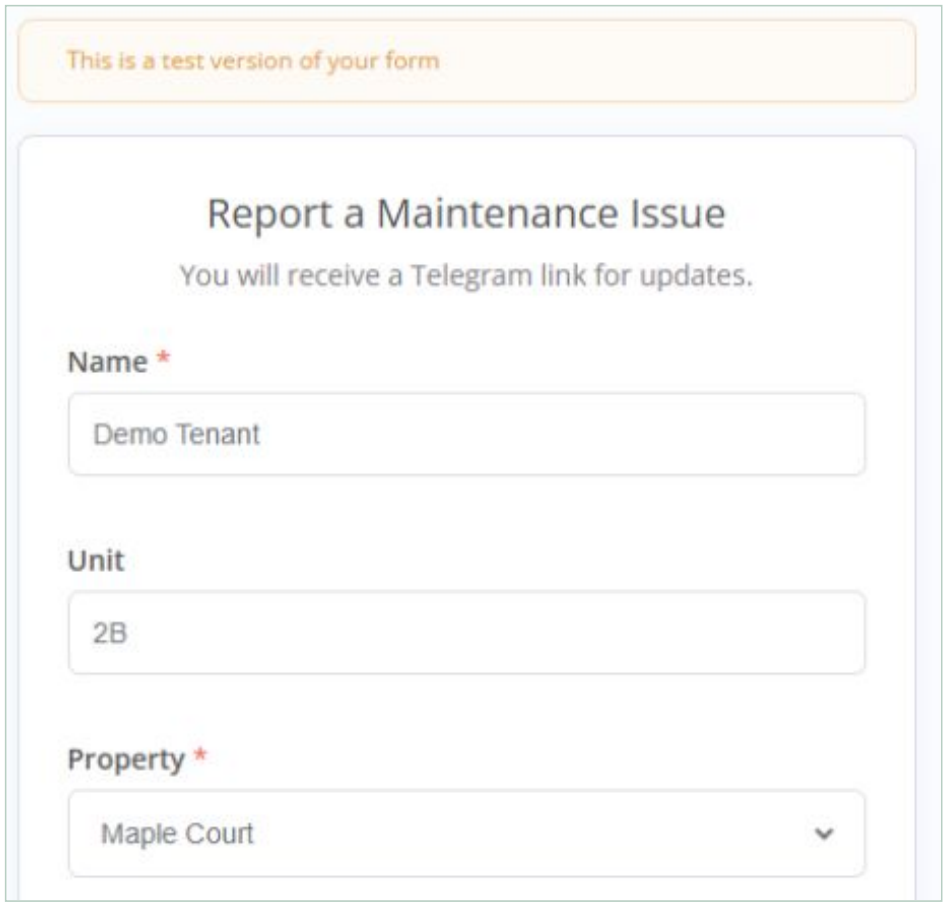


Actual editor capture: enlarged at native resolution.

SMTP sends the Telegram onboarding email to the controlled demo mailbox. The delivery result is attached to the saved request history. A form submission then receives a completion screen; vendor dispatch remains a separate approval action.

A concrete form request

The second demonstration uses the same synthetic tenant and property as the Telegram journey.



Actual form: tenant, unit and property.

The submitted issue

The bathroom exhaust fan stopped spinning yesterday. There is no smoke, burning smell or sparking. No pets are present.

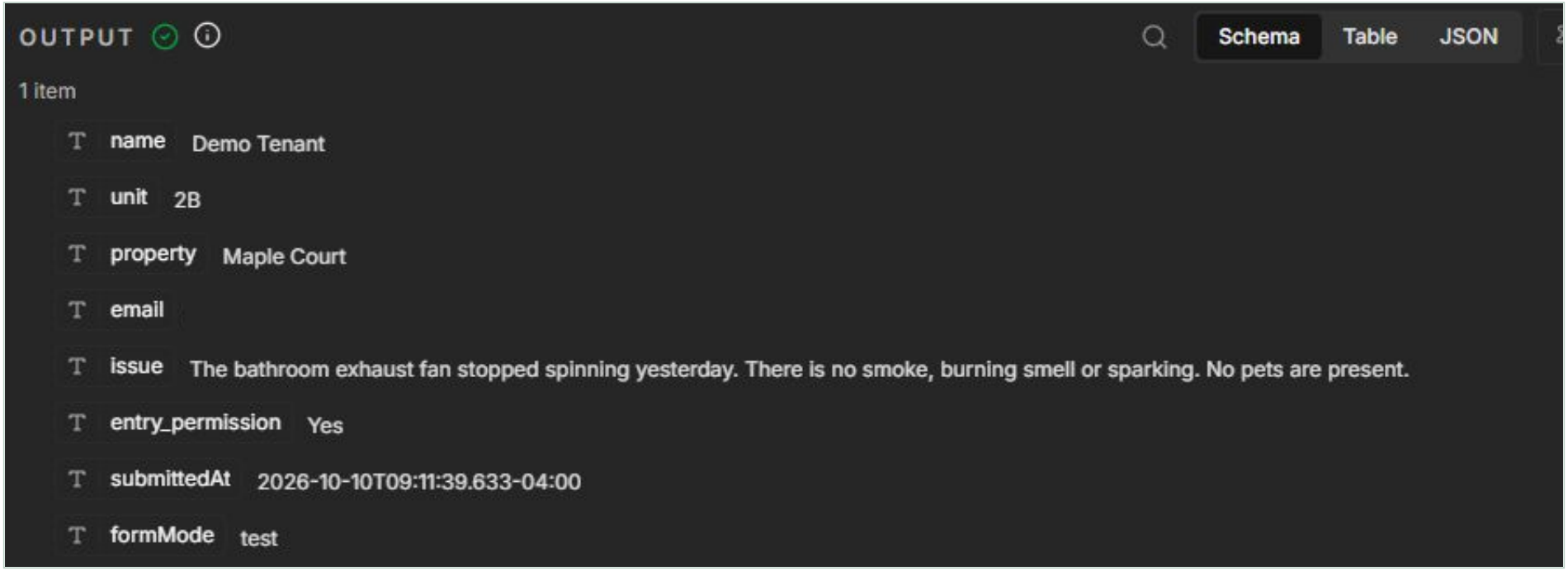
Included context

Demo Tenant / Maple Court, Unit 2B / Entry permission: Yes

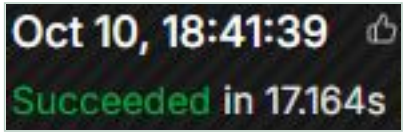
The next page shows the actual submitted payload and completion screen. Contact fields are removed or flattened out of the public captures.

The form became MR-0009

The real submission completed the intake path and created a ROUTINE / APPLIANCE request.



Actual Form Trigger output. Contact value redacted; test mode is visible.



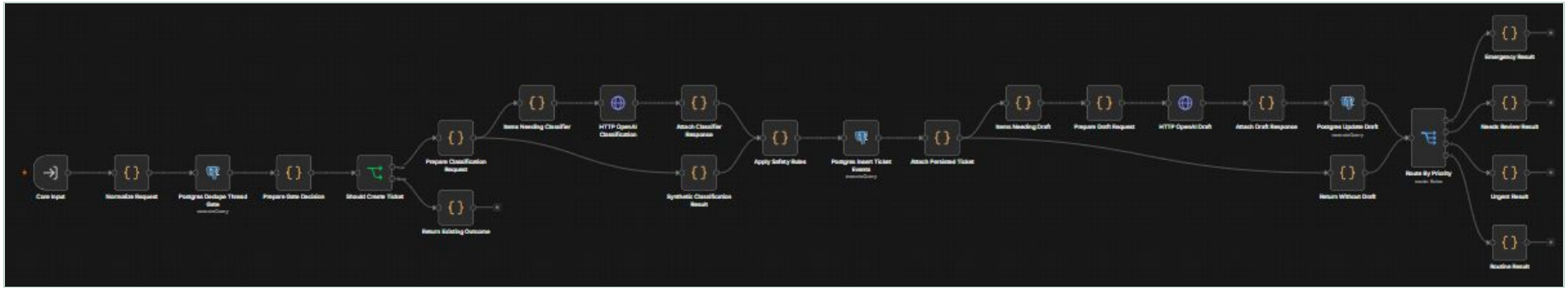
Saved result: MR-0009. Its vendor status remains AWAITING_VENDOR; no vendor email was sent for this request.



Actual receipt after completion.

Interpret, check and persist

The engine is called by intake or tenant conversation flows once there is a concrete maintenance issue.



Actual n8n canvas: every node visible. Readable sections follow.

Incoming data	Normalized message, source identifier, trusted tenant context and test-mode flag.
Decisions	Check prior processing; classify the issue; apply fixed safety and trade boundaries.
State	Postgres retains the ticket, classification reason, events and an eligible vendor draft.
Output	A saved result routed as emergency, needs review, urgent or routine.
Approval	The engine recommends and drafts. The manager chooses whether to contact a vendor.

WORKFLOW 02 / CLASSIFICATION

A structured reading of the issue

The live model identifies request type, summary, priority, trade, access notes and missing information.



Actual editor capture: enlarged at native resolution.

The workflow calls gpt-5.6-luna through OpenAI and requires a structured response. A separate test-mode route can use mock classifications for fixtures; it was not used for MR-0007 or MR-0009.

Check the result before drafting

Fixed rules check safety and maintenance semantics before the ticket event is persisted.

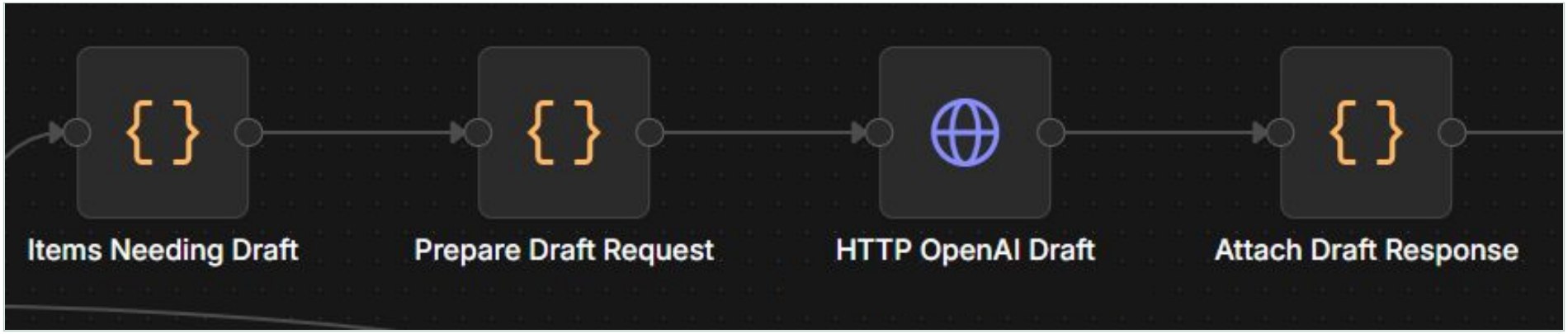


Actual editor capture: enlarged at native resolution.

The saved priority reflects both the structured model response and explicit safety and trade boundaries. Human-only or review-only requests skip automatic drafting; the engine does not use a draft as permission to send.

Prepare a note without sending it

Only an eligible saved maintenance request receives an automatic vendor draft.



Actual editor capture: enlarged at native resolution.

The workflow filters eligible tickets, builds the draft request, calls OpenAI and attaches the returned note. The note is saved for manager review. Preparing it does not dispatch a vendor or promise an appointment.

Four explicit priority routes

The saved result exits through a named priority branch. Downstream workflows use that state to decide the next step.



Priority router.



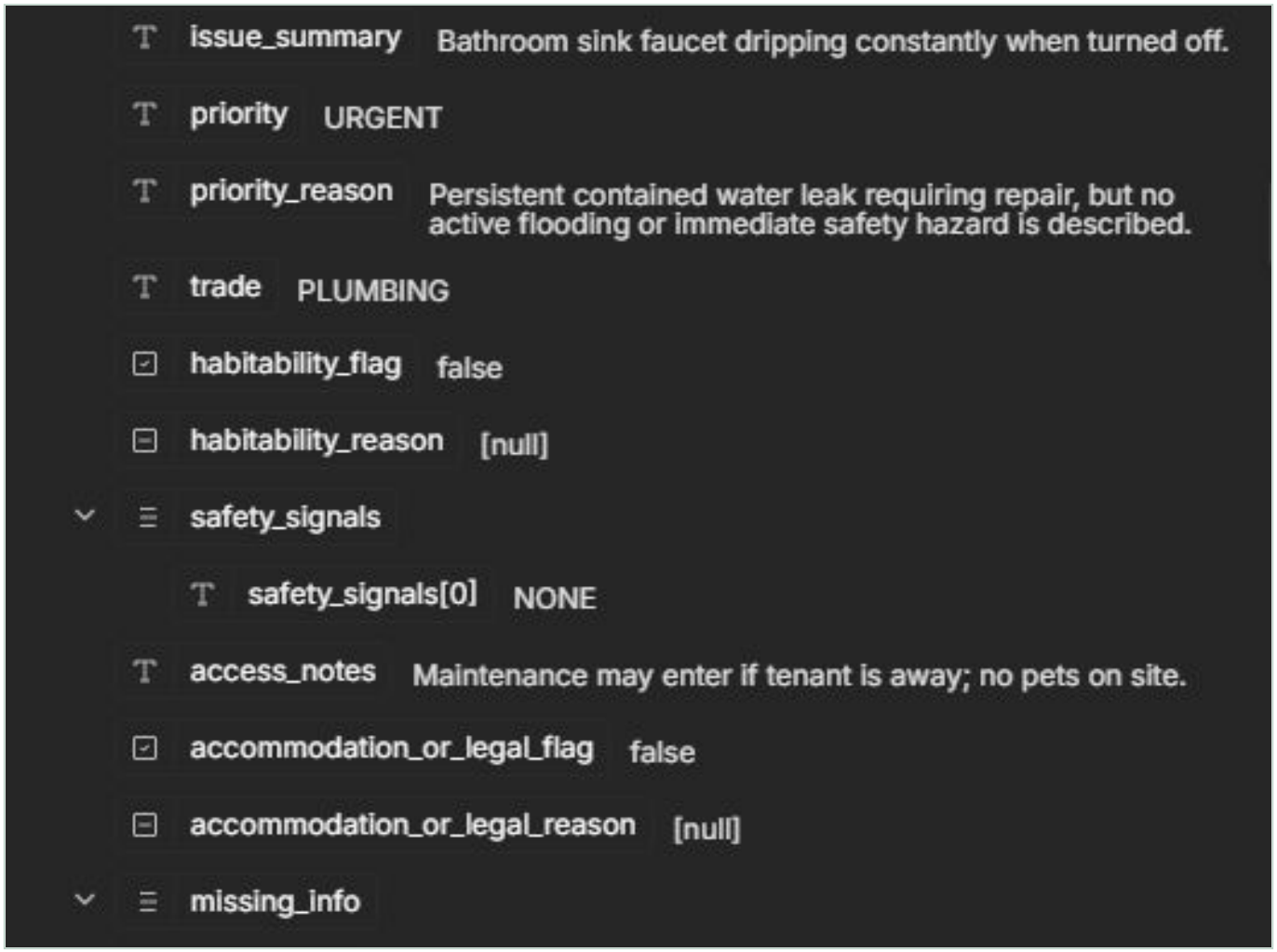
Emergency / review outputs.



Urgent / routine outputs.

MR-0007: a contained faucet leak

This is the recorded live execution after the tenant supplied entry permission and pet information.



Actual checked classification from n8n.

Saved interpretation

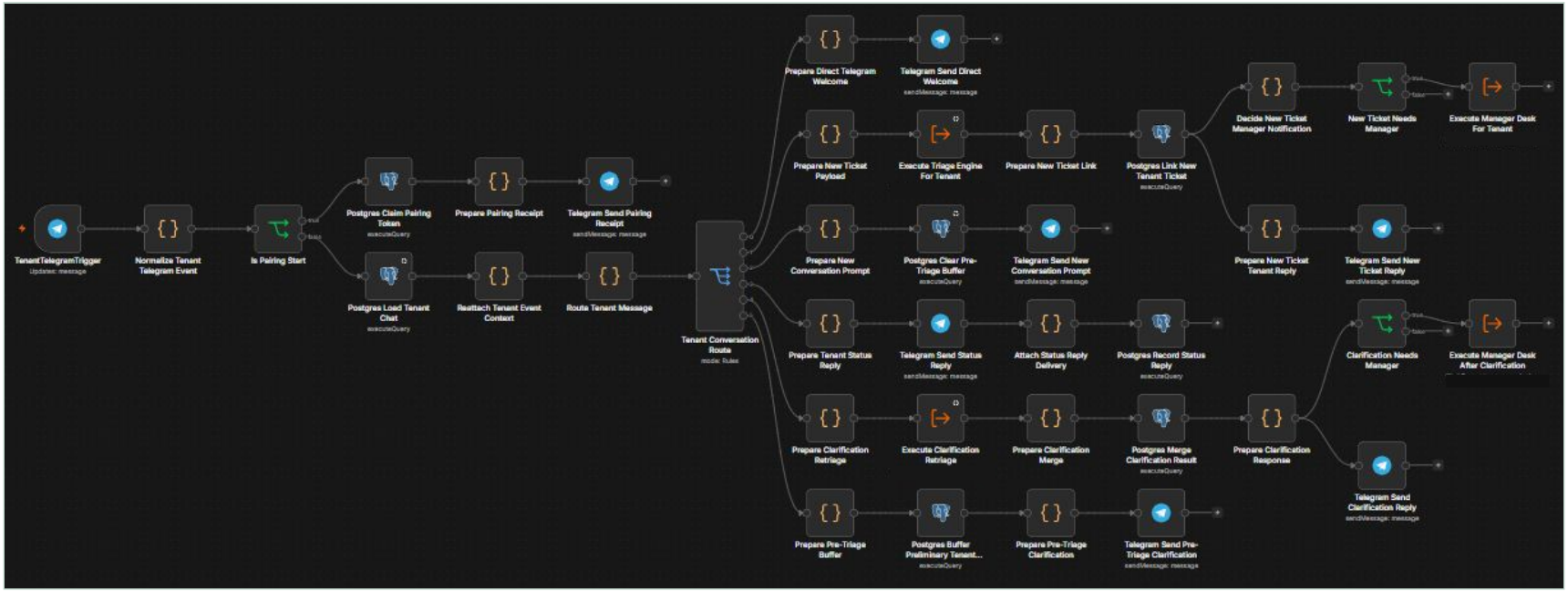
URGENT / PLUMBING

The reason distinguishes persistent contained water from active flooding. The access note allows entry and confirms no pets.

Postgres verification found no remaining missing-information fields. Manager review followed only after this clarification.

Keep the request conversational

A tenant message or pairing link starts this workflow. Saved conversation state determines the appropriate path.



Actual n8n canvas: every node visible. Readable sections follow.

Incoming data	Telegram text, chat context and a configured synthetic tenant profile.
Decisions	Recognize pairing, welcome, new issue, preliminary conversation, clarification or status.
Data movement	Concrete issues call the shared engine; replies merge into the existing request.
Output	A targeted question, factual progress reply or complete request for manager review.

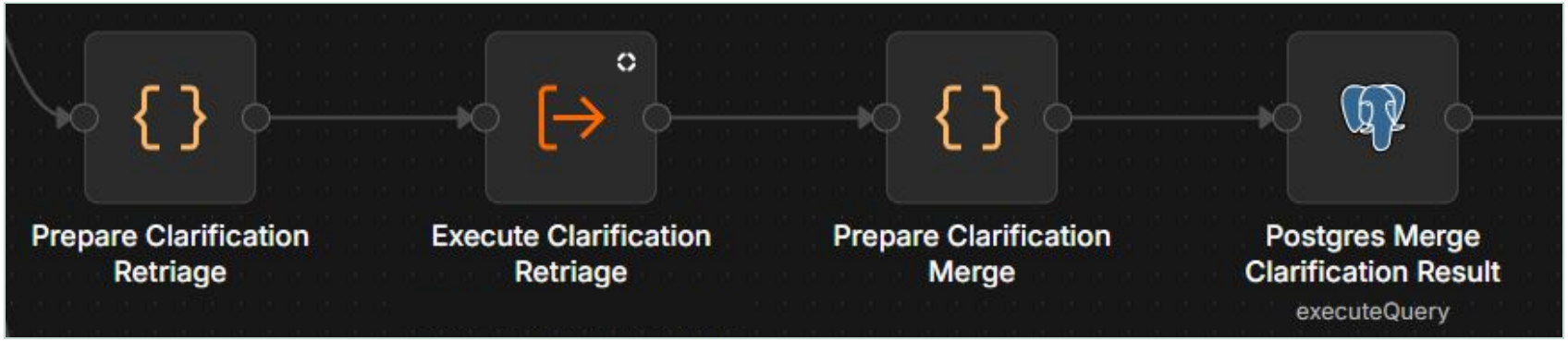
Ask for the detail that matters

A vague opening stays in conversation state. A concrete request with missing information stays in clarification.



Pre-triage buffer: store the opening and ask for a maintenance symptom.

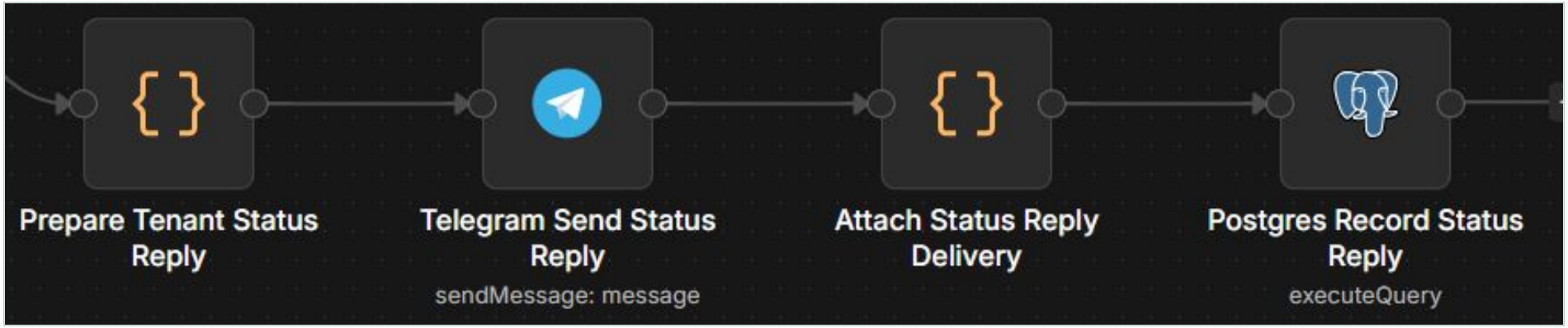
Once a ticket exists, the answer is re-triaged and merged into its original reference. The manager is notified only when the request is complete or explicitly needs review.



Actual clarification route: re-triage and merge into the original request.

Separate progress from new work

Status reads existing tickets. A new issue or clarified request reaches the manager only through an explicit eligibility decision.



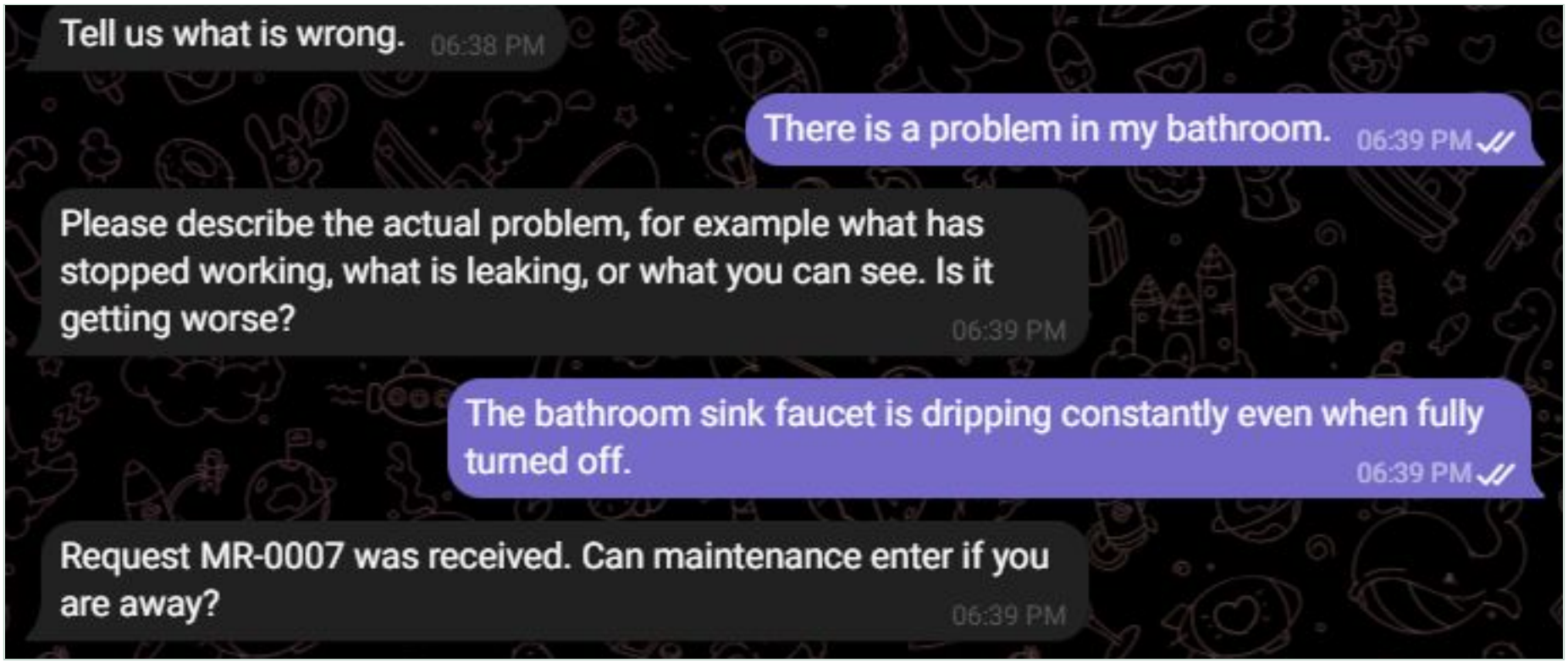
Status route: build a factual reply, send it and record delivery.



Manager handoff: decide eligibility before calling the Manager Desk.

From a vague opening to MR-0007

These are consecutive messages in the active tenant conversation. The synthetic profile supplies Maple Court, Unit 2B.

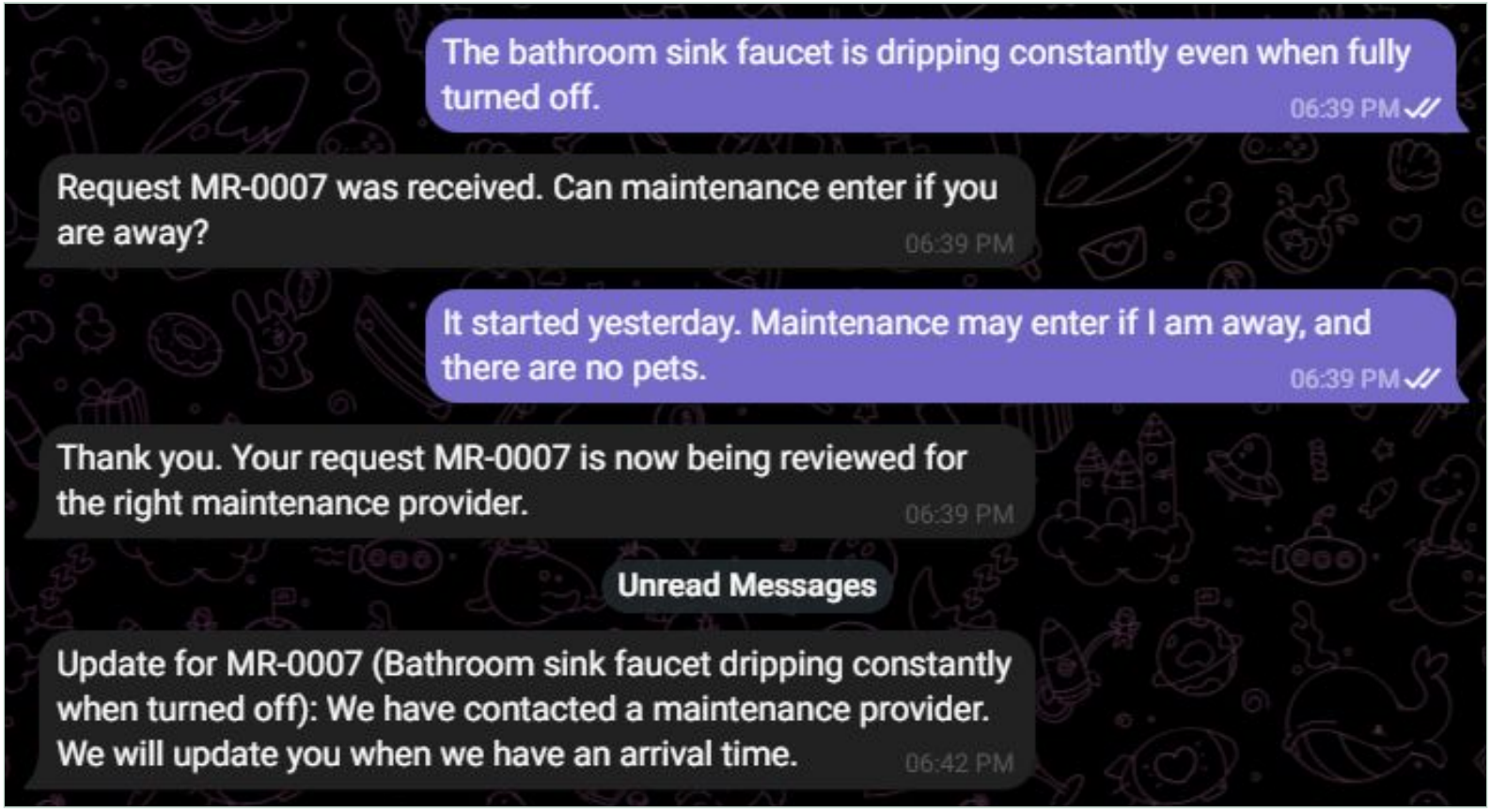


Actual Telegram conversation pane. Other chats and contacts are excluded.

The bathroom opening created no ticket, Airtable record or manager alert. The concrete dripping-faucet message then created MR-0007 and prompted for entry permission. The tenant also supplied timing and pet information in the answer.

One reply unlocked manager review

The tenant confirms when the issue started, allows entry and reports no pets. The same MR-0007 reference continues.

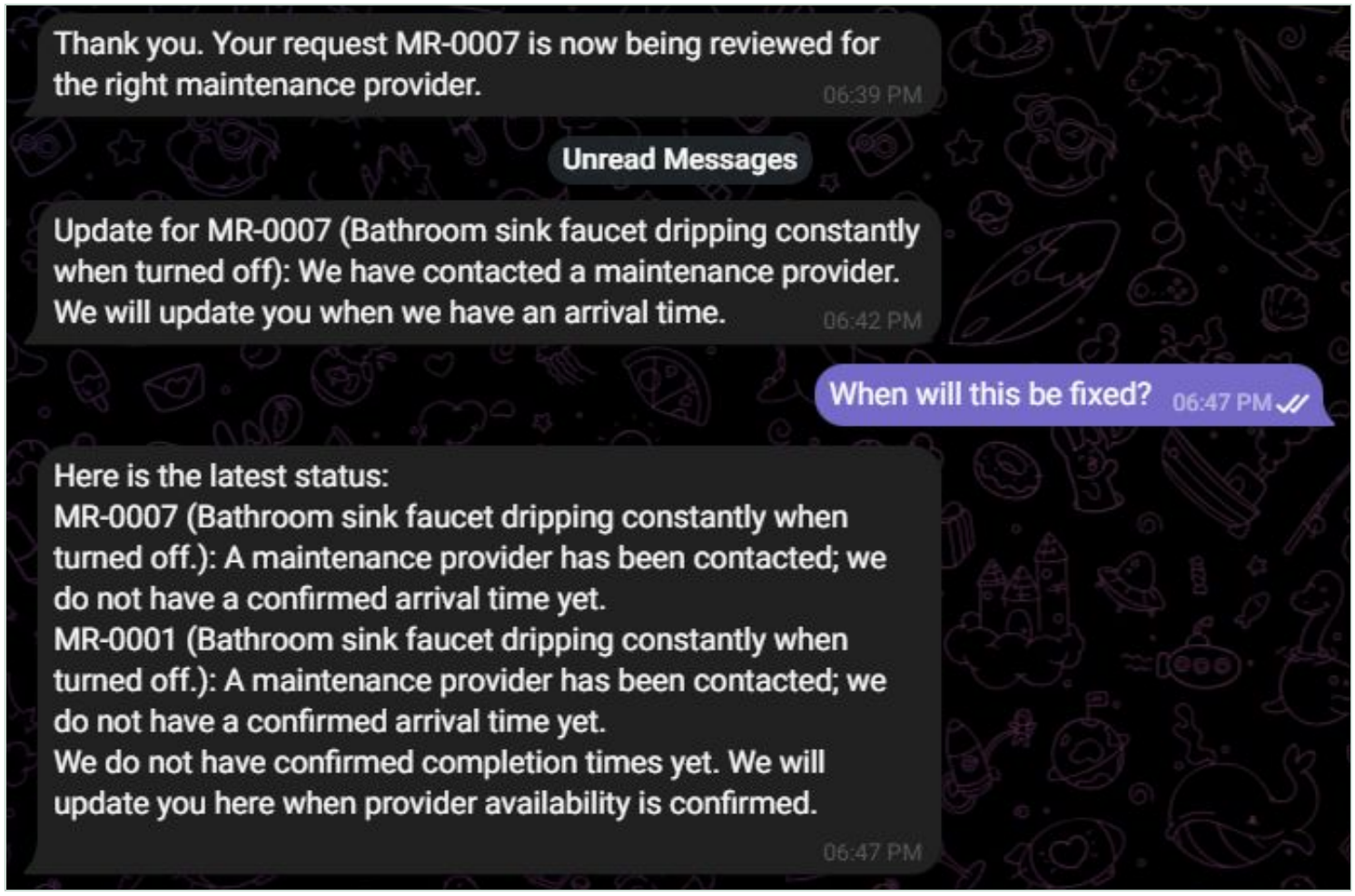


Actual conversation: clarification answer, review notice and provider-contacted update.

The provider-contacted message appears only after the later approval and successful vendor send. It does not claim a booked visit or a completed repair.

A factual answer without a new ticket

The tenant asks when the repair will happen. The workflow responds from open-request state rather than running triage again.



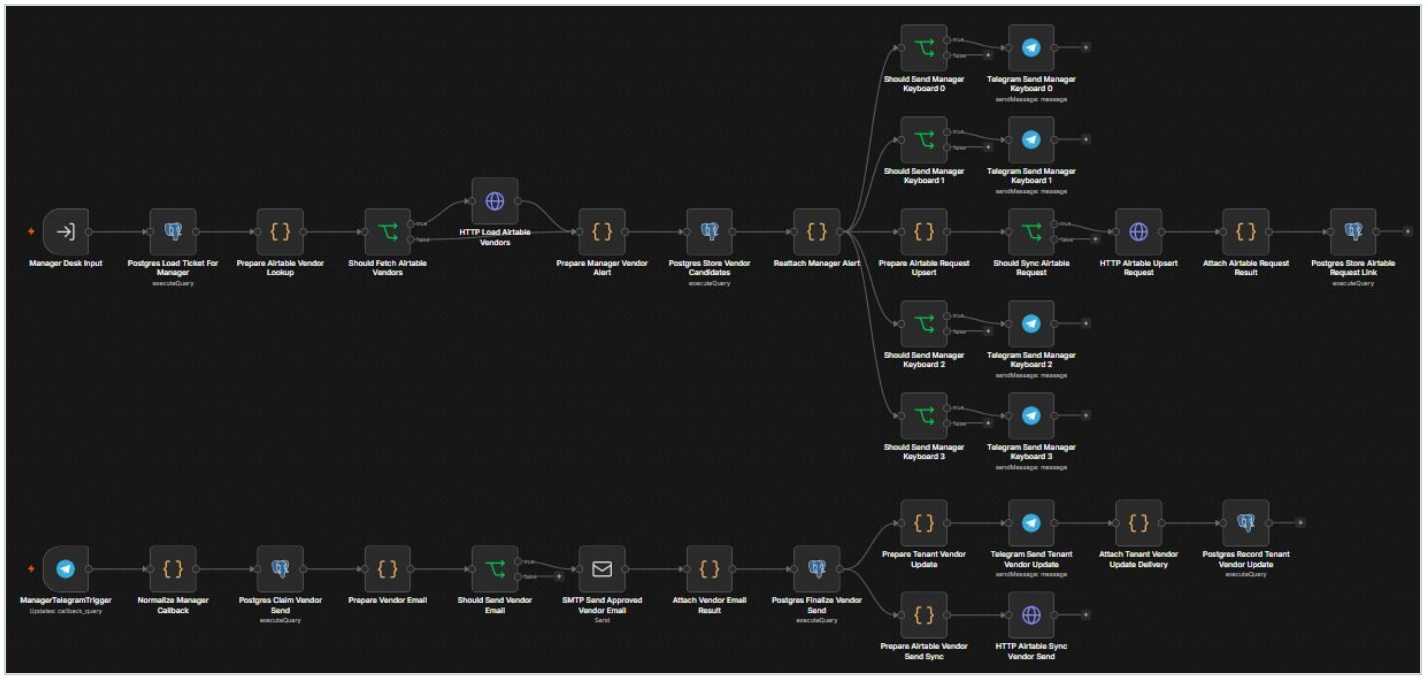
Actual conversation with surrounding messages.

MR-0007 and a preserved open demonstration request, MR-0001, are summarized together.

No arrival time was invented. The database ticket count stayed unchanged, and neither the engine nor manager flow was invoked.

Make the vendor decision explicit

Reviewable requests enter the upper lane. Telegram approval callbacks enter the lower lane.

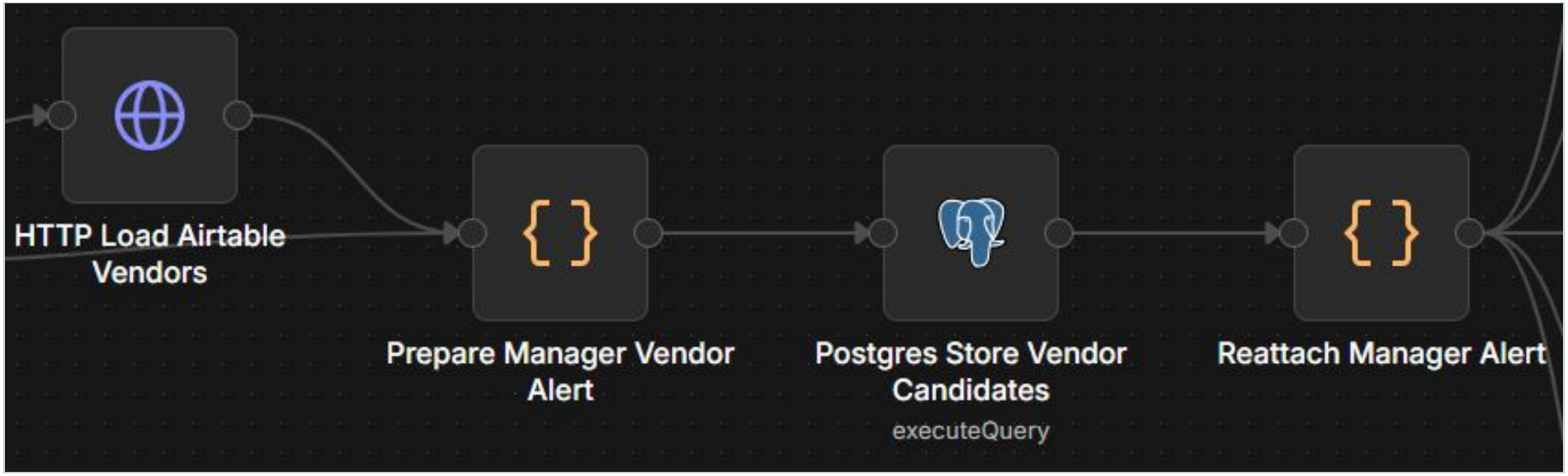


Actual n8n canvas: every node visible. Readable sections follow.

Incoming data	Saved ticket, property, trade, access notes, draft and manager action.
Key steps	Read the vendor roster; save matching candidates; show choices; mirror the request.
Approval	Select a matched vendor or Hold. Review-only requests expose no vendor-send action.
Output	One approved email, saved send state, a tenant update and an Airtable change.

Present relevant vendor choices

The Airtable roster supplies candidates that serve the request property and trade.



Actual editor capture: enlarged at native resolution.

The workflow saves candidates in Postgres before preparing the manager message. It reattaches the request context after the database lookup, so the alert and approval keyboard retain the correct ticket and vendor choices.

Claim one send before using SMTP

A Telegram button supplies the chosen vendor. The stored ticket must permit that action before any email is sent.

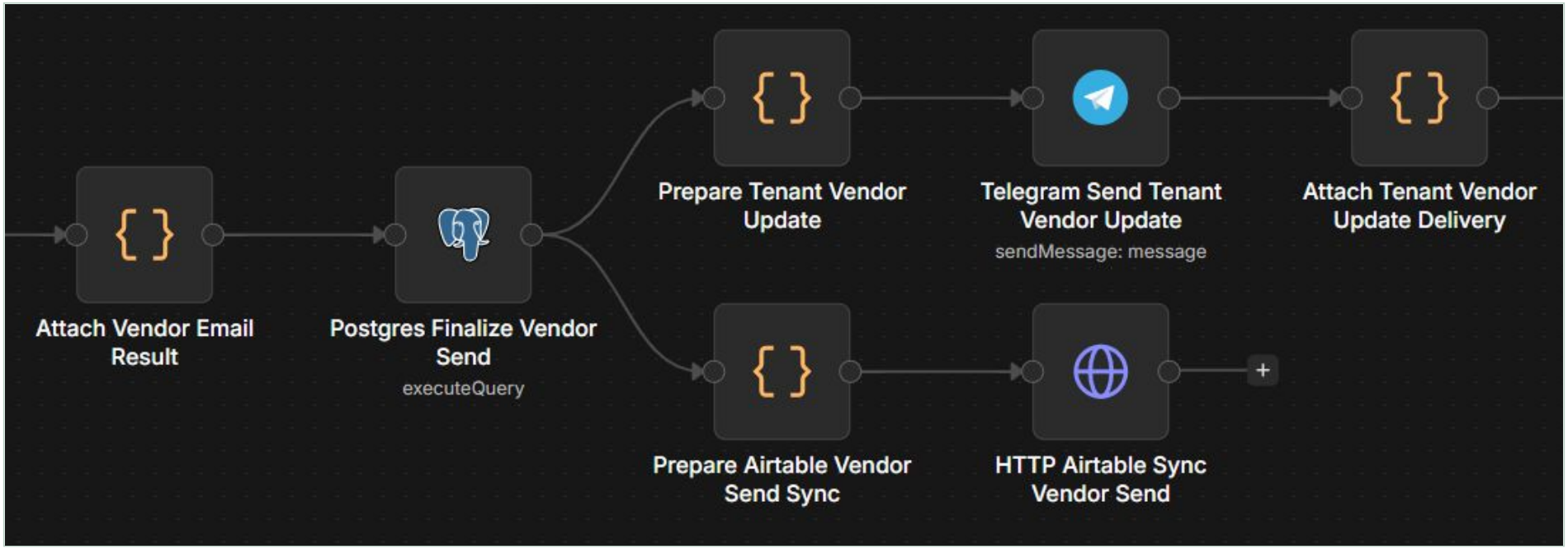


Actual editor capture: enlarged at native resolution.

Postgres claims the send atomically. The email route runs only when the claim permits it, preventing a repeated tap from dispatching the same request again. A Hold or already-handled outcome exits before SMTP.

Update the tenant and the workspace

A successful email is finalized in durable state before the downstream updates run.



Actual editor capture: enlarged at native resolution.

The tenant is told a maintenance provider was contacted. The update delivery is recorded, and Airtable receives the selected vendor and send status. The flow records vendor contact; it does not imply a confirmed appointment.

The prepared request and the action

MR-0007 reaches the manager only after the tenant supplies the missing access detail.

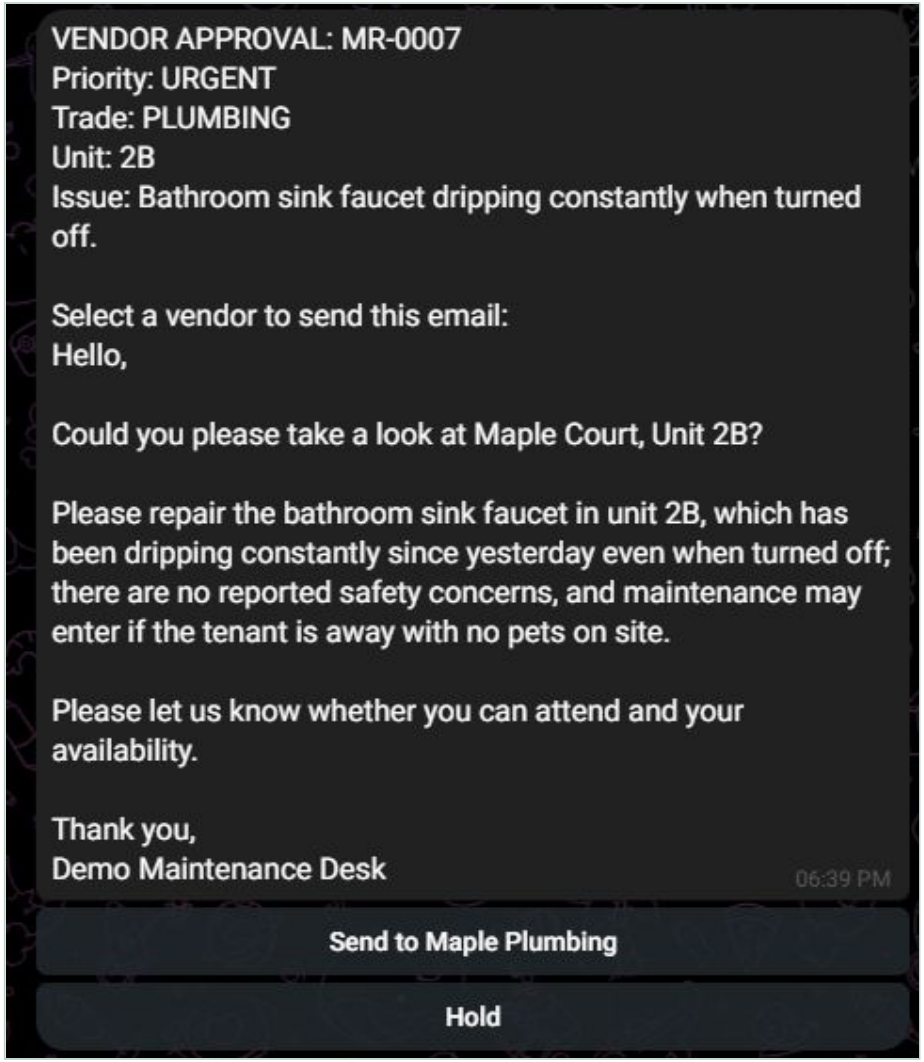
What the manager can review

The message includes the reference, urgency, trade, unit, issue summary, access permission and vendor email draft.

Demonstrated approval

For this controlled demonstration, the operator clicked Send to Maple Plumbing. This exercises the same approval button a property manager would use.

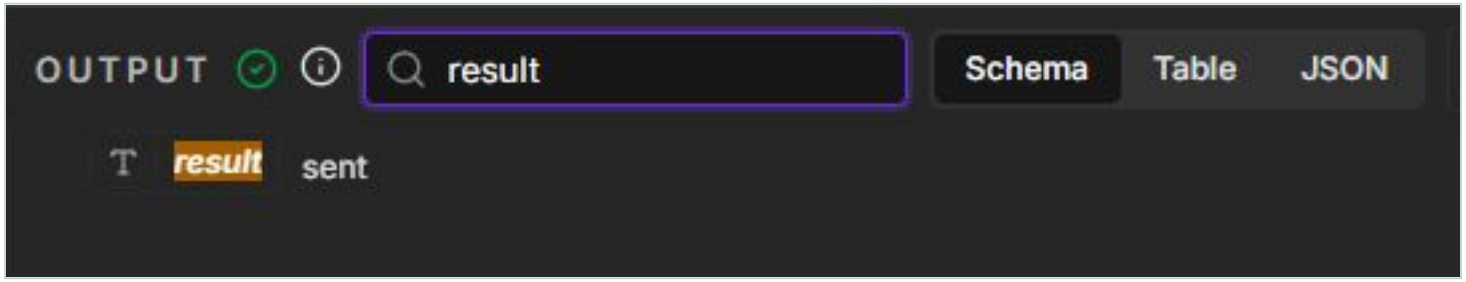
The vendor name is synthetic. Sending used the configured demo mailbox, so the sample shows real delivery without contacting a real contractor.



Actual manager conversation pane.

The approved email was delivered

Postgres finalized the send as SENT. The received subject and body match the faucet request and access details.



Actual n8n finalization result: sent.

[MT-SYSTEM][VENDOR] Maintenance request - Maple Court, Unit 2B Inbox ×

Demo Maintenance Desk 6:42 PM (1 minute ago)

to me

Hello,

Could you please take a look at Maple Court, Unit 2B?

Please repair the bathroom sink faucet in unit 2B, which has been dripping constantly since yesterday even when turned off; there are no reported safety concerns, and maintenance may enter if the tenant is away with no pets on site.

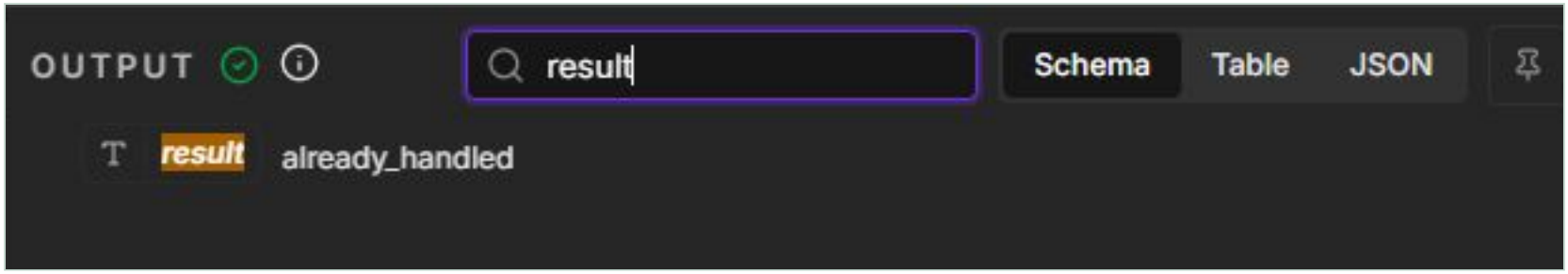
Please let us know whether you can attend and your availability.

Thank you,
Demo Maintenance Desk

Actual Gmail delivery to the controlled demo account. Contact address and sidebars excluded.

Repeated approval stopped safely

The same MR-0007 vendor button was pressed again after the successful send.



Actual n8n claim output: already_handled.

What executed

Callback normalization and the Postgres claim ran. The send decision rejected another dispatch.

What changed

No additional SMTP send ran. The durable record still contained one successful vendor-send attempt.

Both repeated callback executions stopped before SMTP. This evidence pairs the execution decision with the unchanged business state, rather than relying on a green status alone.

Sent and awaiting approval are distinct

The visible references, priorities, sources and linked conversation records were checked against Postgres before capture.

	Reference	Priority	Vendor Status	Selected Vendor
1	MR-0001	URGENT	SENT	Maple Plumbing
2	MR-0002	ROUTINE	NOT_STARTED	
3	MR-0003	ROUTINE	SENT	General Property Repairs
4	MR-0004	URGENT	SENT	Maple Plumbing
5	MR-0007	URGENT	SENT	Maple Plumbing
6	MR-0009	ROUTINE	AWAITING_VENDOR	

Actual Airtable Requests grid with synthetic records.

MR-0007 / faucet leak

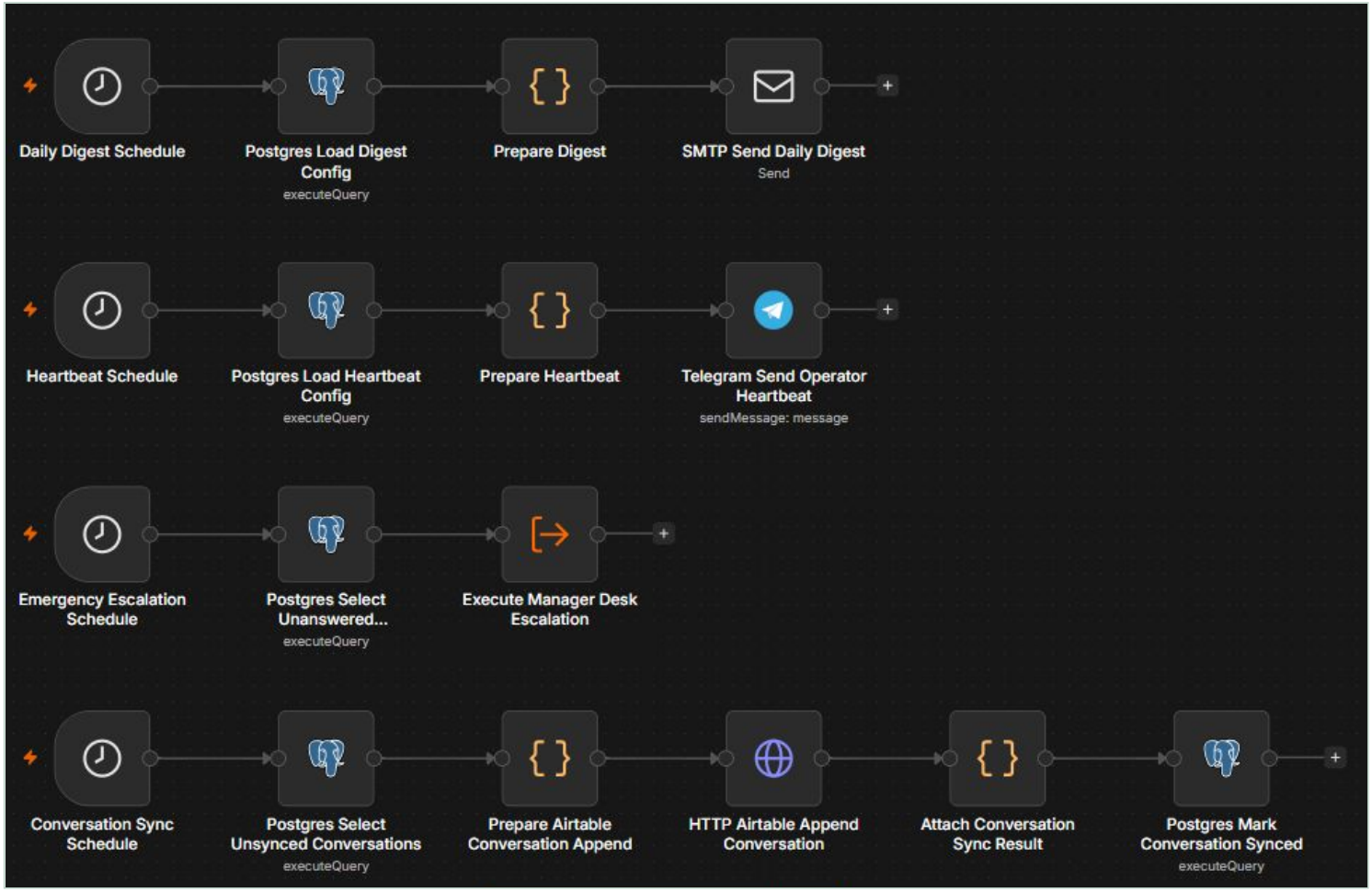
URGENT request; vendor status SENT; selected vendor Maple Plumbing.

MR-0009 / exhaust fan

ROUTINE request; AWAITING_VENDOR. No selected vendor or vendor email yet.

Four separate scheduled lanes

Schedules read saved state. Conversation sync keeps the manager view aligned with the tenant interaction.



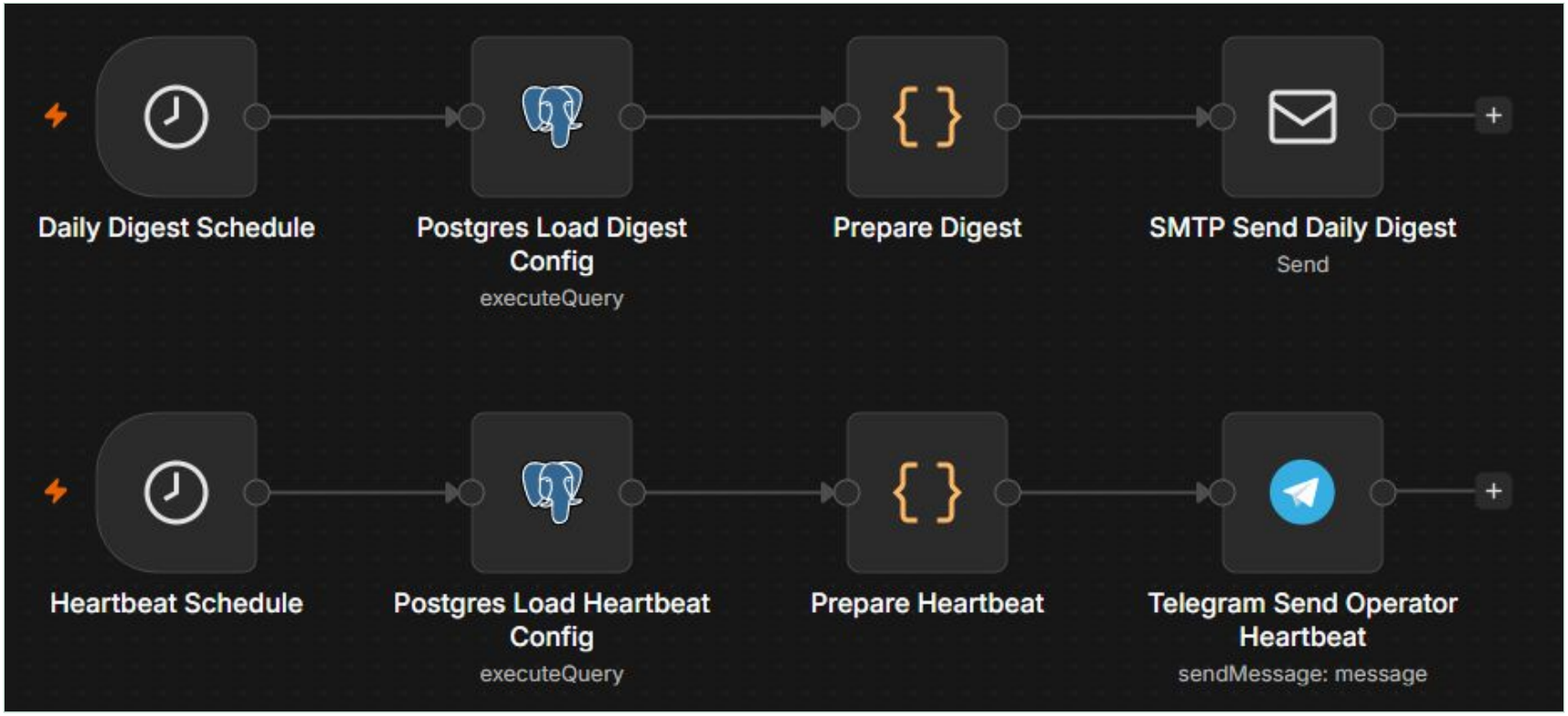
Actual full canvas: all four lanes and every node.

Daily notification / Operator heartbeat / Emergency re-alert / Conversation sync

Each lane has its own trigger and state lookup. Only the emergency lane can invoke manager escalation.

Notifications follow configuration

The daily email and operator Telegram heartbeat load their configuration before preparing any message.

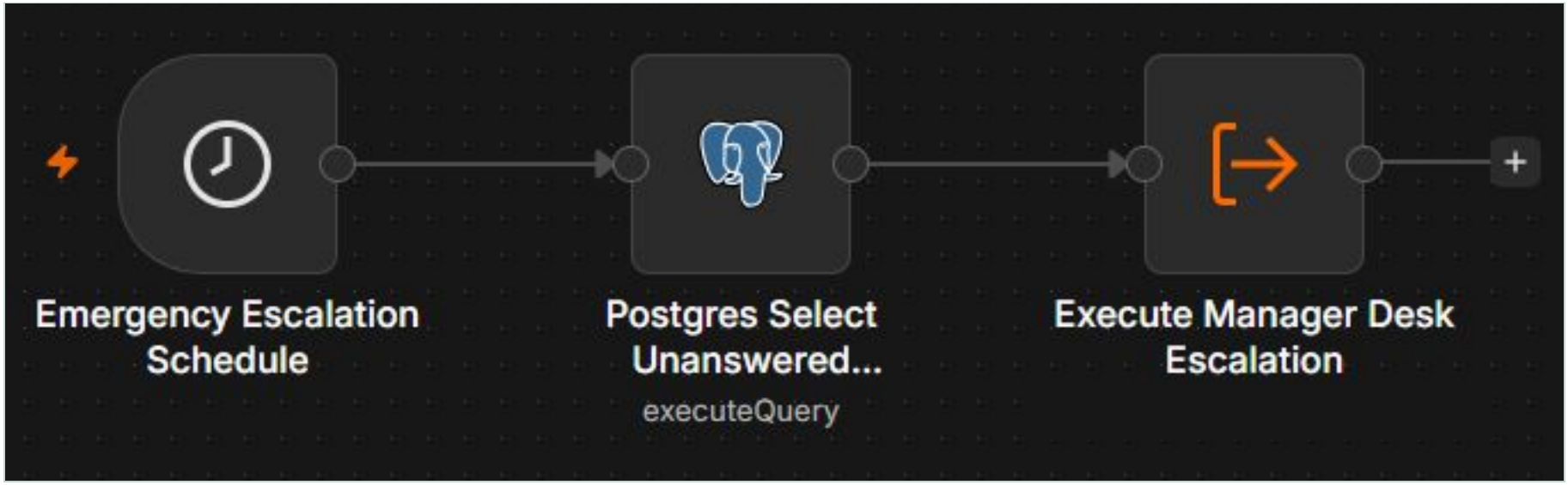


Actual editor capture: enlarged at native resolution.

Both notification features were disabled during this demonstration. The configured daily email is a short reminder to review open requests in Airtable; this sample does not present an unexecuted digest as a delivered output.

Age alone does not create an emergency

The scheduled query selects unanswered emergency requests before re-entering manager review.

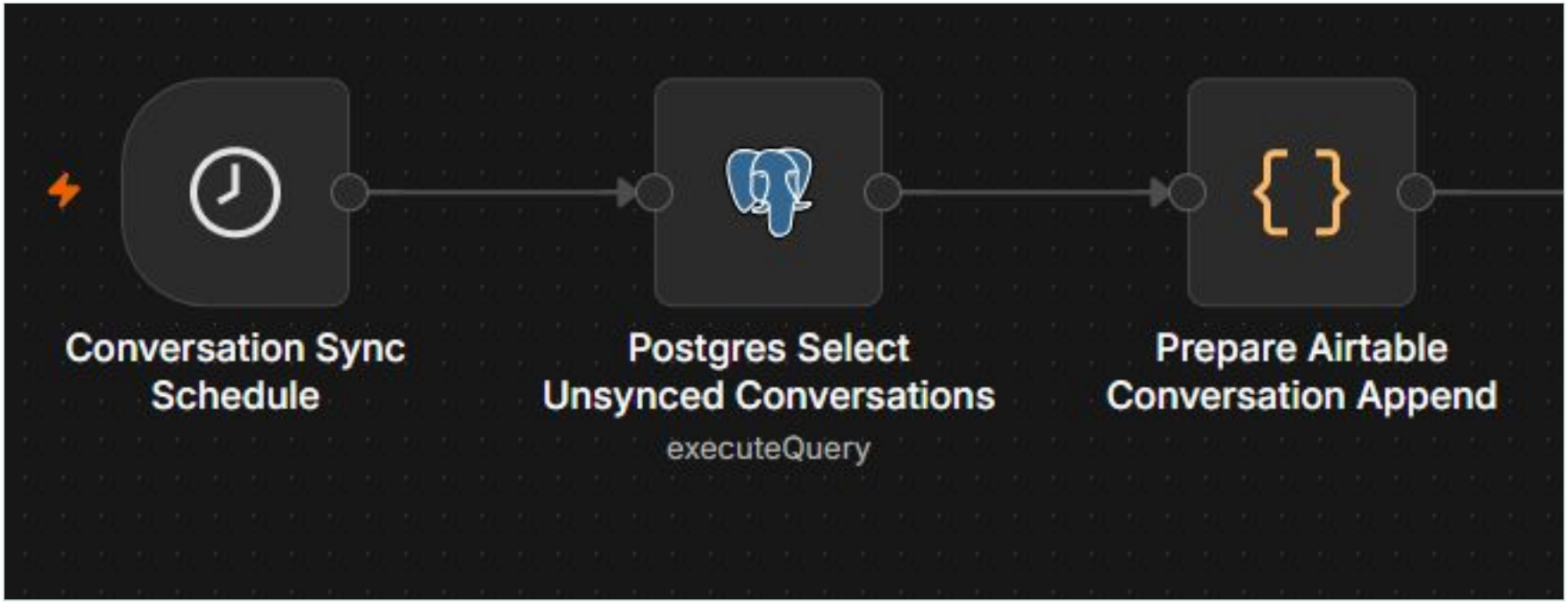


Actual editor capture: enlarged at native resolution.

Only open EMERGENCY requests awaiting vendor handling and older than 15 minutes enter this lane. It calls the Manager Desk for review. MR-0007 remained URGENT and was excluded; the query does not turn request age into an emergency priority.

Select unsynced conversation entries

The five-minute schedule reads messages retained in Postgres and prepares linked Airtable records.

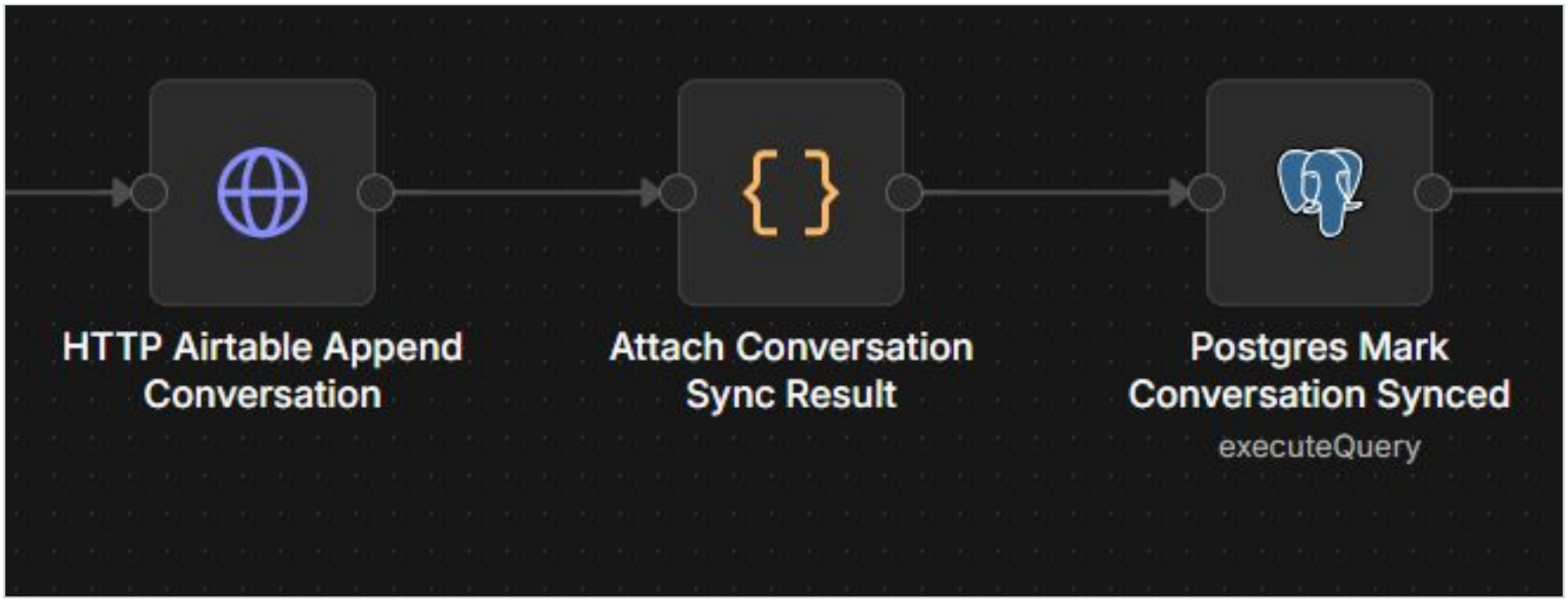


Actual editor capture: enlarged at native resolution.

Each append includes a synthetic conversation label, request reference, direction, channel, intent and delivery state. The same request stays traceable across clarification, vendor updates and status replies.

Append first, then mark completion

The external append result is preserved before Postgres records the message as synced.

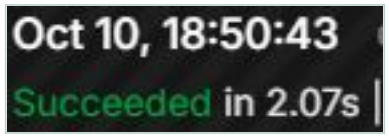


Actual editor capture: enlarged at native resolution.

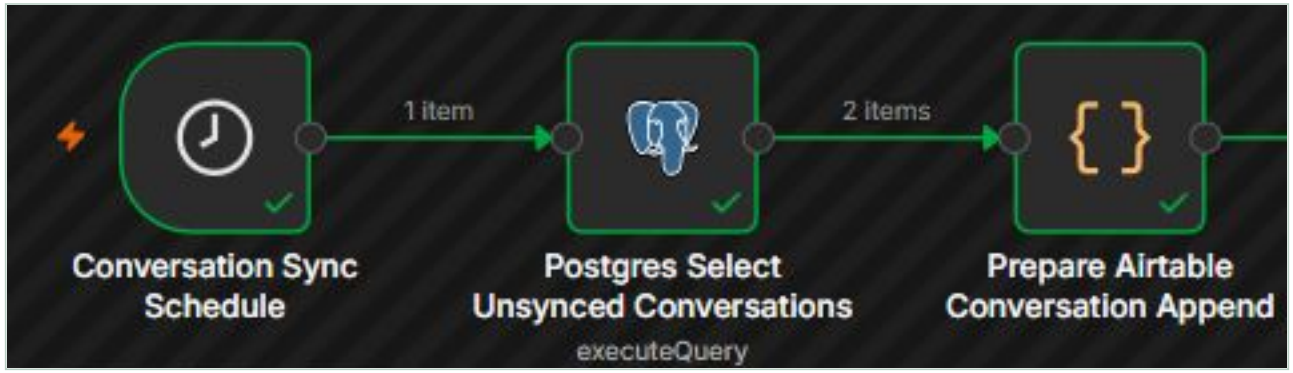
Airtable receives the conversation entry and its request link. The workflow then marks that durable entry as synced. This separates an unsynced message from a completed mirror operation and prevents normal repeated appends.

Status messages reached Airtable

The scheduled run selected and synced two entries from the MR-0007 status exchange.



Actual run: two selected messages, successful Airtable appends, and completed Postgres sync marks.



Executed selection and preparation: two items.



Executed append and sync mark: two items.

Every conversation keeps its request

The actual Airtable conversation view retains reference and direction without exposing message bodies or contact fields.

	Conversation	Request	Direction
1	MR-0001 #38	MR-0001	inbound
2	MR-0001 #39	MR-0001	outbound
3	MR-0002 #40	MR-0002	outbound
4	MR-0003 #41	MR-0003	outbound
5	MR-0004 #42	MR-0004	outbound
	MR-0007 #45	MR-0007	inbound
7	MR-0007 #46	MR-0007	inbound
8	MR-0009 #47	MR-0009	outbound
9	MR-0007 #48	MR-0007	outbound
10	MR-0007 #50	MR-0007	inbound
11	MR-0007 #49	MR-0007	outbound

Actual linked conversation records.

MR-0007 has its request message, clarification answer, provider update and status exchange under the same reference.

MR-0009 retains its email pairing entry. These visible links were checked against the durable records.

Intent and delivery remain visible

This is the adjacent portion of the same conversation grid, in the same row order as the preceding page.

Channel	Intent	Delivery State
telegram	request	RECORDED
telegram	vendor_dispatch	SENT
email	pairing	SENT
email	pairing	SENT
email	pairing	SENT
telegram	request	RECORDED
telegram	clarification_answer	RECORDED
email	pairing	SENT
telegram	vendor_dispatch	SENT
telegram	status	RECORDED
telegram	status	SENT

Actual channel, intent and delivery-state columns.

MR-0007 journey

Inbound request / Clarification answer / Sent provider update / Recorded status question / Sent status response

The recorded states correspond to the actual Telegram conversation shown earlier.

VERIFIED DEMONSTRATION

Evidence beyond the canvas

Fresh synthetic executions on 10 October 2026 were checked against actual messages, durable records and Airtable output.

Vague bathroom opening	Clarification prompt; no ticket, manager alert or Airtable request.
Concrete faucet symptom	MR-0007 created; entry-permission question sent before manager review.
Clarification answer	Original MR-0007 updated; missing information cleared; manager draft shown.
Form fan request	MR-0009 saved as ROUTINE / APPLIANCE; intake receipt displayed.
Approved vendor contact	One delivered demo email; SENT state; tenant and Airtable updated.
Repeated vendor button	already_handled; no second SMTP send; one successful send attempt retained.
Tenant status question	Consolidated factual reply; unchanged ticket count; no re-triage or manager call.
Scheduled conversation sync	Two status entries appended and marked synced; matching Airtable rows visible.

These are demonstrated behaviors with synthetic data and controlled services, not measurements of client savings or production performance.